

Lexical Ontology Persistence, Neural Representation, and Computational Modeling vs. Quantum Linguistics

Abstract

Quantum-inspired approaches to language often model word meaning as a *superposition* of potential interpretations, collapsing to a specific sense only under contextual “measurement.” This analogy has generated elegant mathematical models of ambiguity, contextuality, and non-classical probability in semantics. However, it risks conflating observer-level uncertainty with the ontological status of lexical entities themselves.

This paper proposes an alternative framework: Lexical Ontology Persistence (LOP). According to LOP, words are not mere containers of probabilistic potential; they are structured, historically grounded ontological objects. Each word retains an internal architecture of meaning—shaped by diachrony, morphology, and conceptual structure—that persists independently of any particular context of use. Ambiguity reflects which region of this persistent architecture is accessed, not an indeterminate pre-contextual state.

We develop a **hierarchical** version of LOP that extends to morphemes, graphemes, and sublexical patterns, showing that even letters and phonesthemes can function as syntactic and semantic units because they too possess persistent ontologies. We then sketch how LOP can be implemented in **neural** and **computational** models: as multi-scale, distributed “semantic architectures” in the brain and as structured representational manifolds in artificial systems. Finally, we contrast LOP with quantum linguistics, arguing that quantum formalisms correctly capture contextual dynamics but typically operate only along a **horizontal axis** of interpretive trajectories, neglecting the **vertical semantic hierarchy** that constitutes lexical ontology itself.

Keywords: lexical semantics; quantum linguistics; lexical ontology persistence; neural representation; computational modeling; ambiguity

1. Introduction

Quantum-inspired approaches to language and cognition have attracted increasing attention in recent years. By importing concepts such as superposition, interference, and projection from quantum theory, these models aim to capture empirical phenomena that classical probability struggles with: order effects in judgments, non-distributive reasoning, and context-dependent shifts in interpretation (Aerts 2009; Busemeyer & Bruza 2012; Pothos & Busemeyer 2013). Within semantics, this has led to the idea that ambiguous or polysemous words may be treated as quantum states that collapse onto particular senses when “measured” by context.

The present paper argues that this attractive metaphor misplaces the locus of indeterminacy. It is one thing to model a listener’s or a system’s uncertainty about which sense of *bank* is intended in a given sentence; it is another thing to treat the word *bank* itself as ontologically indeterminate prior to contextual measurement. I propose that words and their sublexical constituents are better understood as **structured, historically grounded entities**: they have internal architectures of meaning that persist across contexts, even when not currently activated.

I call this view **Lexical Ontology Persistence (LOP)**. LOP is not merely a claim about the existence of dictionaries or mental lexicons. It is the claim that each lexical object possesses a rich, multi-dimensional ontology—shaped by the history of the language community and the individual learner—which constrains and guides contextual interpretation. Quantum formalisms, on this view, describe the dynamics of how we move through this ontology; they do not define the ontology itself.

After introducing the superposition model in quantum linguistics (Section 2), I formulate the core LOP thesis (Section 3) and extend it **hierarchically** below the word to morphemes, graphemes, and phonesthemes (Section 4). I then relate LOP to current work in **cognitive neuroscience** (Section 5) and **computational modeling** (Section 6). Section 7 compares LOP with quantum linguistics, including a new distinction between **horizontal** and **vertical** dimensions of lexical ontology. I conclude with directions for empirical, neural, and computational work (Section 8).

2. Quantum Linguistics and the Superposition Model of Meaning

Quantum linguistics and quantum-inspired semantics borrow the formalism of quantum mechanics—Hilbert spaces, superposition, projection operators—to model phenomena such as:

- lexical ambiguity (e.g. *bank* “financial institution” vs. *bank* “river edge”),
- order effects in judgments,
- context-dependent meaning shifts,
- non-classical probability in word association and decision-making (Aerts 2009; Busemeyer & Bruza 2012; Pothos & Busemeyer 2013).

The central picture is familiar. An ambiguous word is represented as a **superposition** of basis states corresponding to different senses, for example:

$$|bank\rangle = \alpha |bank_money\rangle + \beta |bank_river\rangle + \dots$$

Context acts as a **measurement operator** that “collapses” the state onto one sense with some

probability, as in:

- *The bank approved my loan* (financial institution)
- *The bank eroded during the flood* (river edge)

This yields elegant mathematical accounts of context effects and graded interpretation. However, it invites a conceptual slide: because the formalism models meaning as a superposition before context, the *word itself* is often implicitly treated as **ontologically indeterminate**—as if the lexeme *bank* did not have a determinate internal structure until context collapses it.

From the perspective developed here, this is a category mistake. The superposition metaphor correctly describes the listener's or model's **uncertainty** over multiple interpretations; it does not describe the ontological status of the lexical item as a historically formed, structured entity.

3. Lexical Ontology Persistence: Core Thesis

3.1 From surface ambiguity to internal architecture

A word like *bank* has:

- a **diachronic history**: etymological pathways, semantic shifts, borrowings;
- a **morphological profile**: derivatives such as *banker*, *banking*, *bankable*;
- a **conceptual structure**: roles, frames, and typical participants (bank-teller, account-holder, riverbank-slope);
- a **distributional footprint**: collocations and usage patterns across registers and genres.

This is not a shapeless “cloud of potential.” It is a structured internal ontology built up over time.

We can represent this informally as:

$$\text{Arch}(\text{bank}) = \langle \text{H}_{\text{bank}}, \text{C}_{\text{bank}}, \Sigma_{\text{bank}} \rangle$$

where:

- H_{bank} is the historical trajectory of the lexeme (etymology, semantic change, reanalyses);
- C_{bank} is its combinatorial profile (syntactic frames, collocations, affixes);
- Σ_{bank} is its semantic potential space: the structured set of interpretations and conceptual roles it can support.

Crucially, $\text{Arch}(\text{bank})$ exists even when nobody is thinking or speaking about banks, and it remains what it is across different contexts.

3.2 The Lexical Ontology Persistence (LOP) Theorem

We can now state the central claim more generally:

Lexical Ontology Persistence (LOP) – Core Formulation

For any lexical item L in a language, there exists an internally structured ontology $\text{Arch}(L)$, comprising its historical trajectory H_L , combinatorial profile C_L , and semantic potential space Σ_L . This ontology:

1. Persists independently of any particular token, speaker, or context of use.

2. Precedes and constrains contextual interpretation: context does not create Arch(L) but selects, weights, and coordinates regions of Σ_L .
3. Is phylogenetically and ontogenetically layered: it reflects both the community's diachronic history and the individual speaker's acquisition trajectory.

Under LOP, ambiguity is not an ontological superposition of equally unrealized meanings. It is the observer's uncertainty about which already structured region of Σ_L has been selected in the current utterance.

3.3 Bidirectionality: phylogenetic and ontogenetic development

A word's ontology is bidirectionally constructed.

Phylogenetic axis (language history).

Sound changes, semantic shifts, grammaticalization, borrowing, and analogical extension leave durable traces in Arch(L) (Traugott & Dasher 2002). This axis is collective and trans-individual.

Ontogenetic axis (individual acquisition).

Each speaker builds a personal realization of Arch(L) through exposure, usage, and interaction. Frequencies, contexts, and personal experiences shape which subsets of Σ_L are more accessible or salient.

LOP therefore treats each lexical item as a **bidirectional entity**: it is simultaneously the product of long-term community history and ongoing individual learning. Quantum-style superposition models typically operate at the observer level, describing the individual's uncertainty; LOP insists that beneath this there is a **persistent, historically layered architecture** that does not collapse with each context—it is simply evoked differently.

4. Hierarchical Lexical Ontology: Below the Word

LOP becomes especially powerful when extended below the level of the word. Words are not atomic; they are hierarchical lexical objects composed of morphemes, syllables, graphemes, and sublexical clusters. Each of these subunits can exhibit its own ontological persistence.

4.1 From words to morphemes, syllables, and graphemes

Let a lexical object L be decomposed into morphemes:

$$L = \langle m_1, \dots, m_n \rangle$$

Each morpheme m_i can be further decomposed into syllables and segments/graphemes. LOP holds at each level:

- **Word-level LOP:** A lexeme has $\text{Arch}(L)$ as described above.
- **Morpheme-level LOP:** Each morpheme m has its own history, combinatorial behavior, and semantic tendencies.
- **Sublexical LOP:** Syllables, graphemes, and phonesthemes can form semi-stable ontological fields whose histories and usage patterns influence interpretation.

For any sublexical unit u (morpheme, grapheme, cluster), we can analogously say:

Arch(u) = $\langle H_u, C_u, \Sigma_u \rangle$

4.2 Graphemes as syntactic constituents

(a) “L is my favorite letter.”

(1) L is my favorite letter.

Here “L” is a grapheme functioning as the subject of the sentence—a syntactic noun phrase. It is not an ordinary lexical word like *cat* or *table*, yet it:

- bears the topic role,
- enters a copular construction,
- supports predication (“L = my favorite letter”).

This is possible because the grapheme L has its own ontology Arch(L):

- its alphabetic role and order,
- its conventional pronunciation (“el”),
- its history as a letter shape,
- its uses in spelling, abbreviation, and notation.

Context selects the “letter-of-the-alphabet” region of Σ_L , allowing a sublexical unit to skip the usual word level and still function as a meaningful syntactic subject.

(b) “I am here.” / “I is a vowel letter in English.”

(2) I am here.

(3) I is a vowel letter in English.

The form “I” is at once:

- a grapheme,
- a one-syllable phonological form (/aɪ/),
- a free morpheme,
- a lexical word (first-person singular pronoun).

In (2), context activates the pronominal region of Σ_I : “I” denotes the speaker. In (3), context activates the graphemic region: “I” denotes the letter. Under LOP, “I” has a single, layered ontology Arch(I), and different contexts navigate different regions of its semantic potential space.

(c) Letters as grades

(4) She got an A on the exam.

(5) B is the minimum passing grade.

Here “A” and “B” are letters used as noun phrases (object in (4), subject in (5)). Their ontologies include:

- alphabetic identity,
- roles in grading systems,

- evaluative force (A = excellent, B = good).

LOP explains their syntactic behavior by recognizing that each letter is already a stable lexical object with persistent evaluative meaning. The grammar simply accesses Arch(A) and Arch(B).

(d) “There is no I in team.”

(6) There is no “I” in “team.”

Formally, this is about spelling: the grapheme I is absent from the string *team*. Pragmatically, it trades on I as a **pronoun** (ego, self-focus). The line works because a single form “I” carries two overlapping ontologies—graphemic and pronominal—both persistent and culturally entrenched. LOP predicts exactly this kind of layered resonance.

(e) Letters with morphology and idiom

(7) Mind your P’s and Q’s.

(8) Dot your i’s and cross your t’s.

In these idioms, letters:

- take plural morphology (P’s, Q’s, i’s, t’s),
- appear with possessive pronouns (“your P’s and Q’s”),
- function as direct objects of verbs (“mind,” “dot,” “cross”).

They behave like ordinary nouns because each grapheme has an ontology Arch(u) rich enough—alphabetic role, pedagogical routines, writing practices—to sustain **countability** and grammatical relations. Sublexical ontologies are fully recruited by syntax.

(f) X as mark and variable

(9) X marks the spot.

(10) Solve for x.

The grapheme X/x has multiple entrenched roles:

- letter of the alphabet,
- visual mark or cross (on maps, ballots),
- unknown variable in mathematics,
- further uses (e.g. X-rated, Generation X).

LOP treats these as structured branches of Σ_X within a single ontology Arch(X). Sentence (9) selects the “mark” branch; sentence (10) selects the “unknown-variable” branch. This is not quantum-style ontological indeterminacy, but stable polyfunctionality within a persistent architecture.

4.3 Morphemes as persistent ontological units

Consider:

(11) Darkness fell.

(12) Kindness matters.

(13) Marxism is complex.

(14) After Watergate came “Irangate” and “Deflategate.”

Affixes like -ness, -ism, and the reanalysed -gate show clear LOP properties:

- -ness consistently encodes “state/quality of X” (*dark* → *darkness*, *kind* → *kindness*);
- -ism forms ideological or doctrinal nouns (*Marxism*, *feminism*, *surrealism*);
- -gate originates as part of *Watergate* but evolves into a productive morpheme meaning roughly “political scandal involving X.”

For each morpheme *m*, we can similarly write:

$$\text{Arch}(m) = \langle H_m, C_m, \Sigma_m \rangle$$

where H_m is the historical origin and reanalysis trajectory, C_m the combinatorial constraints (which bases it attaches to, in which genres), and Σ_m the semantic tendencies (state/quality; doctrine; scandal).

When speakers interpret “Deflategate”, they are not improvising a brand-new mapping; they are drawing on a historically stabilized ontology of -gate as “scandal,” plus knowledge of *deflate* and football culture. Context selects a compatible region of $\Sigma_{\text{-gate}}$; it does not create it.

4.4 Sublexical phonesthemes and sound–meaning clusters

Between morphemes and individual segments lie phonesthemes and recurrent clusters:

- gl-: *glitter*, *glow*, *gleam*, *glare*, *glimmer*, *glass*, *glimpse*, *glance*
- sn-: *sniff*, *sneeze*, *snore*, *snout*, *snicker*, *snack*

These clusters often concentrate in particular semantic domains:

- gl-: light, vision, shining surfaces;
- sn-: nose, mouth, breath, small noisy actions.

From a purely distributional view, this is a statistical quirk. LOP instead sees sublexical ontological fields:

- Each cluster has a co-occurrence history with certain conceptual domains;
- This history yields a latent semantic orientation—a probabilistic but persistent bias;
- New words coined in the gl- field tend to be interpreted as light/vision-related, and speakers can exploit this in naming and wordplay.

We do not need to elevate *gl-* to full morpheme status. It is enough that Arch(*gl-*) consists of a cluster of words, semantic affinities, speaker intuitions, and poetic exploitability: a clear case of sublexical persistence.

4.5 Hierarchical LOP: revised statement

We can now restate LOP in explicitly hierarchical form:

Lexical Ontology Persistence (Hierarchical Form)

For any lexical object *L* in a language—where *L* may be a word, morpheme, syllable, grapheme, or conventionalized sublexical cluster—there exists an internally structured ontology Arch(*L*) with historical trajectory, combinatorial profile, and semantic potential space. This ontology:

1. Persists independently of context, token, and speaker.
2. Is hierarchically decomposable into the ontologies of its subconstituents, which themselves exhibit LOP.
3. Constrains contextual interpretation, such that context acts as a selector and weight

assigner over Σ_L , not as the creator of $\text{Arch}(L)$.

On this view, words are strata of ontological persistence built out of persistent sublexical parts. Quantum-style models capture how we select among possibilities; LOP describes what is there to be selected.

5. Neural Realization of LOP

How might LOP be reflected in the brain? Contemporary cognitive neuroscience suggests that lexical and conceptual representations are:

- distributed across cortical networks (Pulvermüller 1999, 2005; Huth et al. 2016),
- multi-scale, involving phonological, orthographic, and semantic layers,
- stable but plastic: relatively persistent yet modifiable with experience.

LOP dovetails naturally with this picture.

5.1 Words as stable “architectures” in neural space

A word’s ontology $\text{Arch}(L)$ can be understood as a structured pattern of connectivity and activation:

- Phonological and orthographic circuits encode form;
- Semantic and sensorimotor circuits encode conceptual content, frames, and roles;
- Associative and episodic circuits encode usage-based traces and context patterns.

On this view:

- Σ_L corresponds to a region in high-dimensional neural state space: a manifold of possible activation patterns the system can enter when processing L;
- H_L is reflected in the developmental shaping of those circuits (both phylogenetic constraints and ontogenetic learning);
- C_L is realized in the patterns of co-activation with other words, constructions, and contexts.

LOP claims that this architecture exists prior to and independent of any specific token. When you hear *bank* in isolation, your brain does not construct the entire meaning space from scratch; it re-enters a pre-established region of neural state space whose internal structure has been built by history and experience.

5.2 Hierarchical representation: graphemes to words

The hierarchical LOP perspective aligns with evidence that the brain processes:

- letters and graphemes in visual word form areas,
- phonemes and syllables in auditory and motor regions,
- morphemes and words in temporo-parietal and frontal networks.

Each level has its own:

- representational stability (letter categories, phoneme categories, morphemes),
- combinatorial rules,
- learned associations (letters with grades, morphemes with semantic roles, phonesthemes with conceptual domains).

Under LOP:

- A grapheme like X corresponds to a **recurrent, stable neural pattern** that is reactivated in reading, writing, algebra, and map-symbol contexts;
- A morpheme like -gate corresponds to a **distributed pattern** linking political discourse, scandal schemas, and word-formation routines;
- A cluster like gl- is not an explicit “concept” but a **biased subspace** in phonological–semantic mapping.

The brain does not treat these as ephemeral “superpositions,” but as **persistent structures** that guide context-dependent activation.

5.3 Ambiguity and attractor dynamics

From a neural perspective:

- Ambiguity involves **competition** among partially overlapping neural patterns corresponding to different regions of Σ_L ;
- Context provides **constraints** that favor one attractor basin over others (e.g., financial vs. riverbank senses of *bank*);
- The system resolves toward a **stable attractor state** that fits both the lexical architecture and the contextual input.

Quantum-like formalisms offer an appealing **mathematical metaphor** for this competition and collapse. LOP emphasizes that what collapses is the **state of the cognitive system**, not the **ontology of the word itself**. The neural architecture remains; only its **current activation** changes.

6. Computational Modeling of LOP

What does LOP imply for NLP and computational semantics?

6.1 Limitations of unstructured embeddings

Standard distributional models (word2vec, GloVe, transformer-based language models) represent words as:

- **vectors** in a semantic space,
- with positions shaped by contextual co-occurrence.

These models implicitly approximate something like Σ_L : the cloud of contextual neighbors that define the word's usage. However, they often:

- treat senses as **latent mixture components** without explicit internal structure,
- treat morphology and sublexical patterns as noise or at best auxiliary features.

LOP suggests a shift:

Represent each lexical object L not only as a point or vector, but as a **structured, multi-scale ontology**:

$\text{Arch}(L) = \langle H_L, C_L, \Sigma_L \rangle$,

where H_L , C_L , and Σ_L are themselves partially representable in computational terms.

6.2 Multi-scale, hierarchical representations of Arch(L)

A computational LOP-inspired model would:

1. Encode form and sublexical structure
 - Grapheme-level embeddings (letters, letter n-grams)
 - Morpheme-level embeddings and analyzers
 - Phonesthemic or sublexical cluster embeddings
2. Encode diachronic and ontogenetic information (H_L)
 - Historical corpora to track meaning shifts
 - Speaker- or community-specific adaptation for individual ontologies
3. Encode combinatorial profiles (C_L)
 - Syntactic frames and argument structures (e.g. *bank* takes a human agent and money object in financial uses)
 - Collocational networks and lexical constructions
4. Encode semantic potential spaces (Σ_L)
 - Sense-specific subspaces within a shared embedding region
 - Frame-based or graph-based semantic networks (cf. Jackendoff 2002; frame semantics tradition)

Context then becomes not a creator of meaning, but a **selector / projector**:

- It restricts the model's traversal to subsets of Σ_L consistent with the current sentence, discourse, or task;
- It modulates weights over senses and features that already exist in Arch(L).

6.3 LOP and quantum-inspired NLP

Quantum-inspired NLP models sometimes represent:

- words as state vectors in Hilbert space,
- contexts as projection operators.

These tools can flexibly encode incompatibilities, order effects, and non-commutative aspects of meaning. LOP does not deny their utility, but reframes their interpretation:

- The Hilbert space should be read as a space of cognitive or model states, not as the ontological space of the word itself.
- The state vector before context represents our uncertainty or the model's distribution over a structured Σ_L .
- The projections represent contextual operations that select among pre-existing structured possibilities.

LOP thus encourages hybrid models:

- using quantum-inspired mathematics to model uncertainty and contextual dynamics in the system,
 - while grounding those dynamics in non-quantum, historically structured representations of lexical architecture (multi-scale embeddings, lexical graphs, morphological hierarchies).
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7. LOP vs. Quantum Linguistics: Where the Indeterminacy Lives

7.1 Observer-level merging vs. word-level persistence

Quantum linguistics, at its strongest, focuses on observer-level phenomena:

- A listener facing an ambiguous word genuinely entertains something like a superposition of candidate senses.
- Context acts like a measurement that resolves this superposition.

The danger is to read this as:

■ The word is ontologically in a superposed state until context makes it definite.

LOP reverses this:

- The word is ontologically structured and persistent: Arch(L) is there regardless of context.
- The observer / system is in a superposed state of uncertainty about which region of Σ_L is currently intended.
- Context collapses the epistemic superposition, not the lexical ontology.

7.2 Re-reading classic examples

Take *bank*:

- Quantum view: $|bank\rangle$ is a superposition of financial and river senses; context collapses it.
- LOP view: there is a single Arch(bank) whose Σ_{bank} contains several **structured branches** (financial, riverbank, etc.) with historical and combinatorial scaffolding. The listener's internal state is "in superposition" over which branch of Σ_{bank} is active.

Similarly for letters:

- Quantum view might say: X is in a superposition of "letter X," "unknown variable," and "mark on a map."
- LOP view: X has a single Arch(X) whose branches have stable histories (alphabetic, mathematical, symbolic). The listener is uncertain which is currently relevant; context resolves the uncertainty, not X's ontology.

7.3 Complementarity rather than dismissal

LOP does not reject quantum linguistic tools outright. Rather, it repositions them:

- Quantum formalisms are powerful for modeling contextual dynamics, incompatibility, non-classical probability, and the trajectory of interpretation.
- LOP insists that these dynamics operate over a **non-quantum substrate**: a lexicon of **persistent, hierarchical ontologies** at word, morpheme, and sublexical levels.

The combination yields a richer picture:

- LOP: what is there—structured lexical architectures built by history and usage.
- Quantum tools: how we move through those architectures in real-time interpretation.

7.4 Horizontal vs. vertical dimensions: quantum dynamics as a slice of LOP

So far, quantum-inspired models have been treated as tools for describing contextual dynamics: how interpretations shift, interfere, and “collapse” under different linguistic environments. From the perspective of LOP, however, these contextual dynamics are **not external** to the word; they are themselves a dimension of the word’s ontology.

On the LOP view, every lexical object L possesses at least two orthogonal dimensions:

- a horizontal (X-axis) dimension of contextual trajectories: the patterns of activation and interpretation across episodes of use (discourse events, genres, registers, speaker communities);
- a vertical (Y-axis) dimension of internal semantic hierarchy and history: the layered architecture of sublexical units, morphemes, senses, frames, and diachronic stages.

We can think of a word’s life as consisting of stages that are cycled and folded back onto itself: each new contextual use leaves a trace in H_L and C_L , and these traces gradually reshape Σ_L . The “trajectory of interpretation” that quantum models capture is thus **one horizontal cross-section** through a more complex vertical structure. Quantum formalisms typically operate **within the X-axis**: they track how, at time t , a cognitive or computational system moves between competing interpretive states given a fixed repertoire of senses.

LOP insists that this repertoire is **neither flat nor static**. Vertically, along the Y-axis, we find:

- the hierarchical stratification of graphemes, phonesthemes, morphemes, and words;
- the diachronic layering of earlier and later meanings;
- the nested frames and roles that give structure to Σ_L beyond a mere list of senses.

In other words, quantum-inspired models treat meaning **horizontally**—as a sequence of state transitions across contexts—but largely **ignore the vertical semantic hierarchy** that organizes those states. They work with a relatively unlabeled basis of “senses” or “projections,” treating them as atomic options. LOP, by contrast, treats each “option” as itself the tip of a **vertically extended ontology**: senses arise from, and remain constrained by, a **deep stack of historical, morphological, and sublexical structure**.

From this perspective, quantum formalisms do not so much conflict with LOP as **slice through it**: they describe the **horizontal dynamics** of how a system navigates a pre-existing lexical manifold, while LOP describes the **vertical architecture** of that manifold itself. A complete theory of meaning would need both axes: quantum tools for modeling how interpretations evolve across contexts, and LOP for explaining **what structured semantic material** those interpretations are carved from.

8. Conclusion and Outlook

This paper has articulated Lexical Ontology Persistence (LOP) as a theory of word meaning that:

1. Treats words as **ontologically persistent, historically structured entities**, not as amorphous superpositions.
2. Extends this persistence **hierarchically** to morphemes, graphemes, and sublexical clusters, explaining why letters and phonesthemes can act as syntactic and semantic units.
3. Aligns with **neural evidence** for distributed, multi-scale, stable-plastic representations.
4. Points toward **computational models** that represent lexical items as **structured architectures** over which contextual operations select, rather than as contextless vectors whose meaning is created each time.

5. Offers a conceptual correction to quantum linguistic interpretations by locating indeterminacy in the observer's epistemic state, not in the ontological status of the word.

Future work can proceed along several lines:

- **Empirical:** corpus and psycholinguistic studies testing hierarchical LOP—for instance, the productivity of *-gate*, the semantic clustering of *gl-*, and the syntactic behavior of letters such as X, A, and I.
- **Neuroscientific:** imaging and neurophysiological studies probing the persistence of sublexical and lexical architectures across contexts.
- **Computational:** multi-scale models that incorporate Arch(L)-like structures and comparison with quantum-inspired models on tasks involving ambiguity and context.
- **Philosophical:** further clarification of the metaphysical status of lexical ontologies—whether they are abstract objects, distributed patterns, or hybrid entities.

What LOP ultimately proposes is a shift in where we locate “mystery” in meaning: not in the word’s being somehow undefined until observed, but in the **complex, historically sedimented architectures** that words—and their parts—already possess, and in the **dynamic, sometimes quantum-like ways** in which our cognitive systems navigate those structures.

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